Unit Test Report

Railway Reservation System - Team 15

1. Purpose and scope

This report records the execution of the automated unit and integration tests for the Railway Reservation System, as required by Part-3 of the mini-project. Results below were produced by building and running the tests on the final main branch.

Field	Value
Project	Railway Reservation System (RRS)
Team	Team 15 (TEAM15-RRS)
Repository	TEAM15-RRS/se-miniproject-railway
Code version tested	Branch main, commit 7ff1f9c (3 Nov 2026)
Report date	11 November 2026

2. Test environment

Item	Value
OS	Ubuntu 24.04
Compiler	GCC 13.3.0, C++17
Build system	CMake (Debug build)
Test framework	Project's custom framework (tests/test_framework.h), run through CTest
Command	ctest --test-dir build --output-on-failure

3. Summary of results

Test target	Total	Passed	Failed	Pass rate
Unit tests (unit_tests)	141	141	0	100%
Integration tests (integration_tests)	12	12	0	100%
Performance benchmark (search_performance_test)	1	1	0	100%
CTest targets	3	3	0	100%

Total automated test cases: 153 (141 unit + 12 integration), plus 1 performance benchmark. No tests were skipped.

4. Unit tests by file

Test file	Area covered	Tests
test_admin.cpp	Admin module: train add/modify/deactivate, fare/quota rules, admin access, admin log	22
test_admin_reports.cpp	Admin booking and revenue reports	3
test_auth.cpp	Registration, login, password hashing, lockout	13

Test file	Area covered	Tests
test_auth_extra.cpp	Per-user lockout, password length limit, admin login edge cases	4
test_backup.cpp	Backup and restore	4
test_booking.cpp	Booking, PNR, seat decrement, RAC/Waitlist, validation, receipt, PNR lookup	15
test_booking_dev.cpp	Booking on journey day, independent dates, fare, booking time	4
test_booking_extra.cpp	Advance window, per-passenger fare, class code, name trimming	6
test_cancel.cpp	Cancellation, refund slabs, promotion, regression after admin changes	14
test_cancel_search_extra.cpp	PNR formatting, requester case, freed seats, search edge cases	5
test_invalid_input_fuzz.cpp	Invalid-input and randomized fuzz tests for 5 services	5
test_inventory.cpp	Seat inventory and queue positions	6
test_payment_dev.cpp	Card validation, transaction IDs, masking	8
test_receipt_card.cpp	Masked card on receipt	2
test_search_payment.cpp	Train search and payment service	11
test_storage.cpp	File round trip, corrupt lines, temp files, seed data	6
test_storage_extra.cpp	Missing folder, Windows line endings, inactive flag	4
test_util_sha.cpp	SHA-256 and date/time utilities	9
Total		141

5. Integration tests

ID	Scenario	Modules involved
IT-01	Register, save, reload, login	Auth + Storage
IT-02	Password never written in plaintext	Auth + Storage (RRS-SR-001)
IT-03	Lockout survives restart	Auth + Storage (RRS-SR-005)
IT-04	Booking persists and search reflects it	Search + Booking + Storage
IT-05	No PNR collision after reload	Booking + Storage
IT-06	Cancelled PNR not reused after reload	Booking + Cancel + Storage
IT-07	Waitlist promotion across restart	Booking + Cancel + Storage
IT-08	Failed payment leaves seats and files untouched	Booking + Payment
IT-09	End-to-end register, search, book, cancel, refund	All services
IT-10	Abrupt termination leaves data readable	Storage (RRS-NF-003)
IT-11	Foreign user cannot cancel	Auth + Cancel
IT-12	Corrupt record does not block startup	Storage (RRS-NF-002)

6. Performance test (RRS-NF-001)

search_performance_test searches a database of 5,000 trains (500 matching routes) and requires every search to finish within 1,000 ms.

Run	Median / Maximum	Result
Team run, CMake Release (Google Colab)	0.819 ms / 0.948 ms	PASS
Re-run for this report, Debug build, 1 CPU	36.6 ms / 44.7 ms	PASS

Both runs are far below the 1,000 ms limit.

7. Requirement traceability

Requirement	Automated tests	Result
RRS-F-001 Registration	TC_Reg_01..04, TC_Fuzz_01	Pass
RRS-F-002 / F-003 Login	TC_Auth_01..04, 08..11	Pass
RRS-F-010..012 Search	TC_Search_01..09	Pass
RRS-F-020..025 Booking	TC_Book_01..27, TC_Pay_01..08, IT-05, IT-08	Pass
RRS-F-030..034 Cancel / PNR	TC_Cancel_01..17, IT-06, IT-07, IT-11	Pass
RRS-F-040..043 Admin	TC_Admin_01..18, TC_AdminReports_01..03	Pass
RRS-F-050 / F-051 Storage and backup	TC_Store_01..10, TC_Backup_01..04, IT-01, IT-04	Pass
RRS-NF-001 Performance	search_performance_test	Pass
RRS-NF-002 / NF-003 Robustness	TC_Store_03, TC_Fuzz_01..05, IT-10, IT-12	Pass
RRS-SR-001 Password hashing	TC_Sec_01, 02, IT-02	Pass
RRS-SR-002 Masking	TC_Sec_03, TC_Pay_06, TC_Book_26 (CLI masking checked manually)	Pass
RRS-SR-003 Admin access control	TC_Auth_04, TC_AdminAccess_RejectsPassengerActions	Pass
RRS-SR-004 Admin log	TC_AdminLog_01..03	Pass
RRS-SR-005 Lockout	TC_Sec_05..07, TC_Auth_08, IT-03	Pass

8. Observations and defects

- No unit or integration test failed on this commit.
- The manual system test of 31 Oct 2026 (docs/SYSTEM_ACCEPTANCE_TEST_REPORT.md) recorded three failures: registration fields (DEF-02), berth preference (DEF-03) and the admin log timestamp (DEF-01). The unit tests passed because they check the implemented

behaviour, not these CLI prompts. These defects must be retested in the CLI before they are marked closed.

- Backup/restore and admin reports now have automated tests (TC_Backup_*, TC_AdminReports_*), which were pending at the time of the Sprint 2 traceability matrix.

9. Conclusion

All 153 automated unit and integration tests pass and the search performance benchmark meets its limit. The unit and integration test objectives of the Test Plan are met on this commit.